

# **Hemodynamic Disorders and Diuretics**

Integrated Block 2 — Session 14  
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# Edema

Edema is an accumulation of interstitial fluid within tissues. It results from increased hydrostatic pressure, reduced plasma oncotic pressure, lymphatic obstruction or sodium retention.

Congestive heart failure is the most common cause of increased hydrostatic pressure. Nephrotic syndrome causes edema through loss of albumin in the urine.

# Virchow triad

Three factors predispose to thrombosis:

1. Endothelial injury
2. Abnormal blood flow (stasis or turbulence)
3. Hypercoagulability

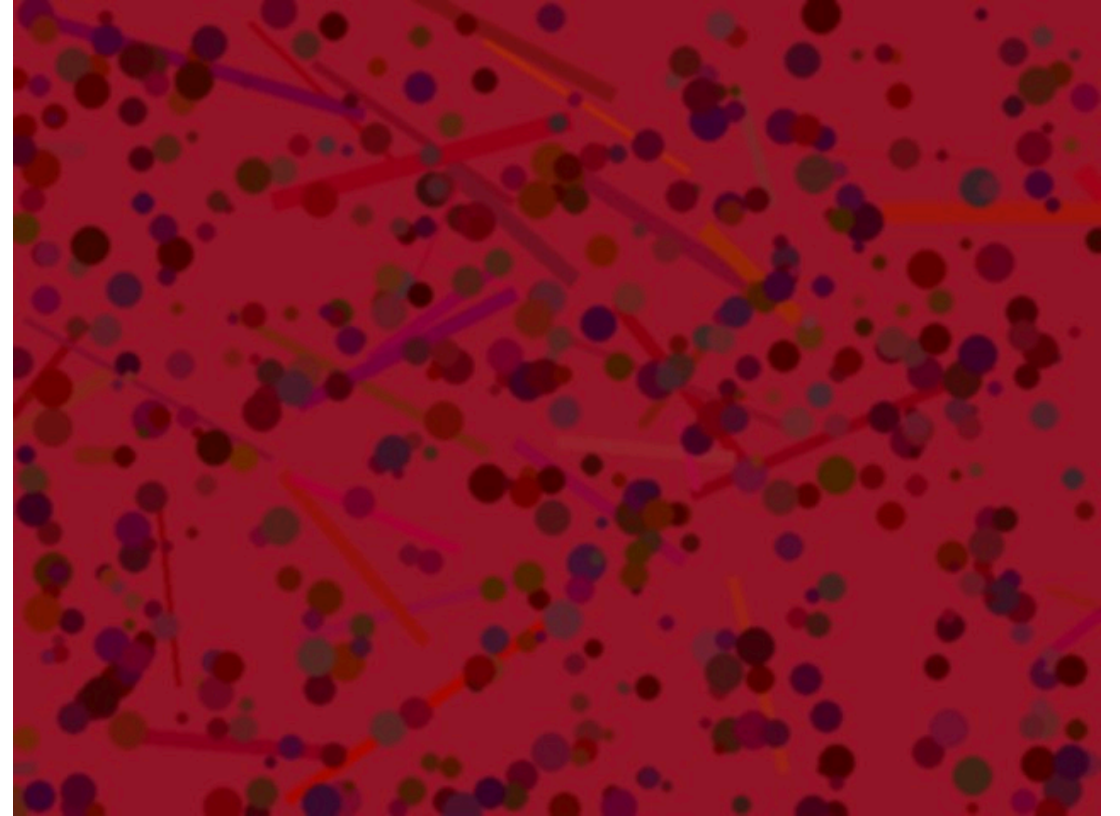
Endothelial injury is the dominant influence in the heart and arterial circulation. Stasis is the major factor in the development of venous thrombi.

# Arterial and venous thrombi

Figure 4: Arterial thrombus with lines of Zahn



Figure 5: Venous thrombus (phlebothrombosis)



Arterial thrombi usually begin at sites of endothelial injury. Venous thrombi almost invariably occlude the lumen. They most commonly involve the veins of the lower extremities.

# Embolism

## **Pulmonary embolism**

More than 95% of pulmonary emboli originate from deep vein thrombi of the legs. A large embolus lodged at the bifurcation of the pulmonary artery is called a saddle embolus.

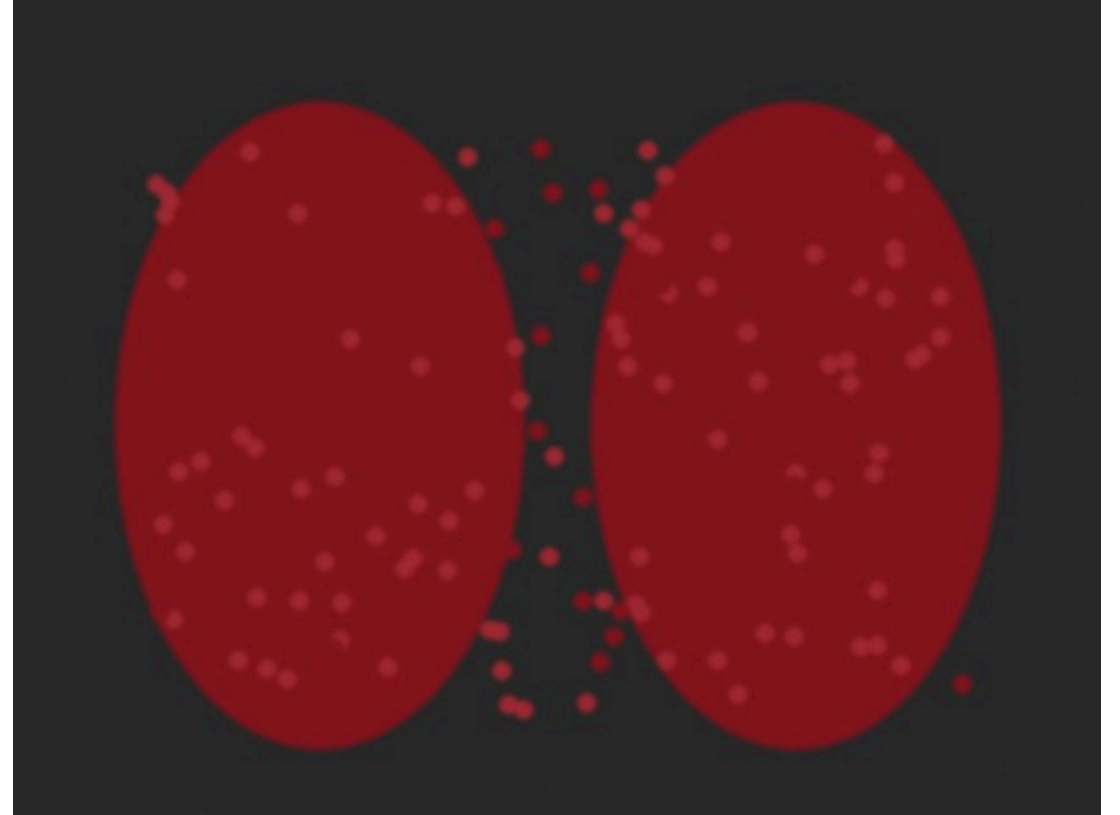
## **Fat embolism**

Fat embolism usually follows fractures of long bones. It causes pulmonary insufficiency, neurologic symptoms, anemia and thrombocytopenia.

# Infarction



White infarct of the spleen



Red infarct of the lung

White infarcts occur with arterial occlusions in solid organs with end-arterial circulation. Red infarcts occur in tissues with a dual blood supply, such as the lung.

# Shock

| Type of shock | Mechanism                                    | Clinical example        |
|---------------|----------------------------------------------|-------------------------|
| Cardiogenic   | Myocardial pump failure                      | Myocardial infarction   |
| Hypovolemic   | Loss of blood or plasma volume               | Hemorrhage              |
| Septic        | Peripheral vasodilation and pooling of blood | Gram-negative infection |

Why does septic shock cause warm, flushed skin?

# Loop and thiazide diuretics

## **Furosemide**

Furosemide inhibits the Na-K-2Cl cotransporter in the thick ascending limb of the loop of Henle. It is used to treat acute pulmonary edema. It can cause hypokalemia and ototoxicity.

## **Hydrochlorothiazide**

Hydrochlorothiazide inhibits the NaCl cotransporter in the distal convoluted tubule. It is used to treat hypertension. It decreases urinary calcium excretion.

# Potassium-sparing diuretics

- Spironolactone is a competitive aldosterone receptor antagonist.
- It is used in primary hyperaldosteronism and heart failure.
- Amiloride blocks epithelial sodium channels in the collecting duct.
- Both drugs can cause hyperkalemia.

| Diuretic      | Site of action                      |
|---------------|-------------------------------------|
| Acetazolamide | Proximal convoluted tubule          |
| Mannitol      | Proximal tubule and descending limb |
| Furosemide    | Thick ascending limb                |

# Summary and questions

- Thrombosis → embolism → infarction
- Know your diuretics!

Any questions?